3 Customizing a Gaussian process with the mean and covariance functions

This chapter covers

- Controlling the expected behavior of a Gaussian process using mean functions
- Controlling the smoothness of a Gaussian process using covariance functions
- Learning the optimal hyperparameters of a Gaussian process using gradient descent

In Chapter 2, we saw that the mean and covariance functions are the two core components of a Gaussian process (GP). Even though we used the zero mean and the RBF covariance function when implementing our GP, you can choose from many options when it comes to these two components.

By going with a specific choice for either the mean or the covariance function, we are effectively specifying prior knowledge for our GP. Incorporating prior knowledge into prediction is something we have to do with any Bayesian model, including GPs. Although I say we "have to" do it, being able to incorporate prior knowledge into a model is always a good thing, especially under settings in which data acquisition is expensive like Bayesian optimization.

For example, in weather forecasting, if we'd like to estimate the temperature of a typical day in January in Missouri, we won't have to do any complex calculation to be able to guess that the temperature will be fairly low. At the other end of the spectrum, a summer's day in California will be relatively hot. These rough estimates may be used as initial first guesses in a Bayesian model, which are in essence the model's prior knowledge. If we didn't have these first guesses, we would have to do more involved modeling in order to produce predictions.

As we will learn in this chapter, incorporating prior knowledge into a GP can drastically change the model's behavior, which could lead to better predictive performance (and eventually more effective decision-making). Using no prior knowledge should only be done when we have

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